Improving Opioid Security and Medication Adherence in the Home

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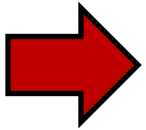
Problem: Lack of Pill Security in the Home

Anyone has access to pills to abuse, misuse, divert and accidentally overdose

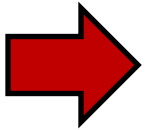
Problem: Lack of Patient Medication Adherence

75% of patients do not take their prescribed medication correctly



$290 billion added cost / year
125,000 unnecessary deaths

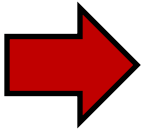
With opioids and opioid use disorder (OUD)



$2.7 trillion added cost / year
112,000 overdose deaths

Problem: Opioid Use Disorder (OUD) = Addiction

Starts in a few days to a couple of weeks …

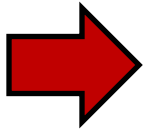


Lasts a lifetime

86% of Heroin addicts report 1st abusing prescription opioids as teens.

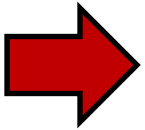
Problem: Patients cannot get treatment for OUD

We have medications to treat OUD



Suboxone Reduces Death Rates by Greater than 50%

But only 11% of patients with OUD can access treatment



Many Barriers to Care

- Lack of prescribers
- Pharmacy closures
- Lack of transportation
- Lack of insurance
- Stigma
- Geographical disparities
- Healthcare illiteracy
- Social Inequity

2 million patients "are in" OUD treatment
27 million patient "need" OUD treatment

Result: Doctors Fear Prescribing Opioids

Doctors fear losing their license because no safeguards to prevent patients from :

Opioid abuse
Opioid Use Disorder (OUD)

Opioid diversion

Accidental overdosing

[>89,111 opioid overdose deaths in 2023]

Solution: Bring Care to Patients with Pain and OUD Safely

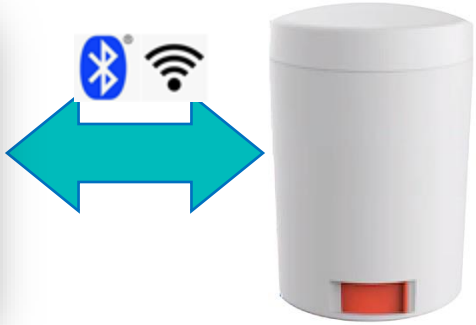
"Mobile banking app" + **"Bank ATM"** → **"For Pills at home"**

Only the prescription holder has access to pills
Only at the prescribed dose and time

The Solution: iPill Dispenser

"**Mobile banking app**"

Biometric Key



"**Bank ATM**"

Drug Vault

iPill App

Monitor pills
Precise dispensing
Remote medication adherence
Alerts patient family, friend, caregiver of missed daily doses for remediation
Monitor Patients
Telehealth for Psychosocial support
Vital sign capture via faceprint

iPill Dispenser

Monitor, secure, destroy pills
Secure home delivery and storage
Tamper-resistance prevents abuse, diversion, accidental overdoses
Destruction of unused pills at 30 days and immediately if tampered

iPill Commercial Production Readiness

Prototype	**POC device completed. Commercial pilot completed**
Samples	**Samples ready for distributionn and sales**
Funding	**Raised $2.2 seed** **Raising $3.25 for manufacturing and commercialization**
Regulatory	**FDA Class I Registered 510K exempt** **FDA Class II Breakthrough Designation**
IP protection	**3 granted US Patents 10426707, 10588824, 10709643** **3 PCT Patents EU, UK, Canada**
Commercialization	**Distribution: UPS Healthcare Services** **Fulfillment: Medvantx Inc.** **Manufacturing: Foxconn (Mexico)**
Commercialization Timeline	**Jan 2025 Foxconn Proof of Concept device complete** **July 2025 Foxconn Mass produce devices complete** **Sept 2025 Commercial launch**

Who pays for it? Not patients, Not Insurance

iPill can sell to multiple clients who pay upfront to create multiple revenue streams:

- Jails
- OUD clinics
- Veterans Benefit Administration
- Specialty pharma
- Contract research organizations (CROs)

iPill avoids dealing with insurance denials, delays and preauthorization

iPill Revenue Model #1 Jails

TAM 6.89 million lives annually = $41.34 billion at $500/iPill

6.89 million detainees (65% of the 10.6 million detainees released per month have OUD)

Jails need

- Detainees released from prison have a 129 higher chance of death in the 2 weeks after release
- Mandatory detainee use of iPill for 12-18 months as part of early parole for OUD treatment

Status with iPill

- iPill has been paid $250 for 10 units for a commercial pilot
- Jails now want iPill units to dispense 120 pills of Suboxone which will raise cost of iPill to $500 per month
- iPill paid upfront with opioid litigation settlement funds

iPill Revenue Model #2 OUD Clinics

TAM US 27.5 million lives annually = $115.5 trillion at $350/iPill

US 27.5 million people have OUD in the US

OUD Clinic need

- Financially tight OUD clinics welcome a revenue share opportunity
- Reduced medicolegal and regulatory liability, reduced workload, increased patient volume, improved care, and improved profitability

Status with iPill

- Commercial pilot with 5 patients completed.
- The OUD clinic is set to order iPill

Provider testimonials: Easy to prescribe iPill; Improved patient monitoring; Protects patients from harm

Patient testimonials: Easy to use; Can I use it again for next month?

iPill Revenue Model #3 Veteran Benefit Administration VBA

TAM VBA 418,000 lives annually = $4.514 billion at $900/iPill

Veterans Benefit Administration Need

- Veterans who could not get care can receive care at home
- 208 clinics treat 6,600 veterans for OUD
- 418,000 lives need treatment but cannot get treatment
- 44 veterans die a day from suicide or overdose

Status with iPill

- iPill is now on the Advanced Procurement Portal
- iPill is in process of FedRAMP, FIPS, and FISMA certification
- VBA is waiting for iPill to begin production
- Payment upfront from Federal Gov't

iPill Revenue Model #4 Specialty Pharma

TAM SPECIALTY PHARMA

67 million prescriptions annually = $80.4 billion at $1200/iPill

Specialty Pharma need

- Each pill for transplants, HIV, or cancer may run $4,000 to $20,000
- insurers cannot monitor if patients take the pills and thus prevent more healthcare costs
- Specialty pharma already covers other $1200 dispensers with far fewer features

Status with iPill

- A group of 10 specialty pharma in Atlanta is set to place order once iPill begins production
- iPill gets paid upfront by the specialty pharma who then charges insurance for reimbursement

iPill Revenue Model #5 CROs

TAM CROs 920,000 lives annually = $1.104 billion at $1200/iPill

920,000 study participants in sponsored clinical trials annually

CROs need

- Remote patient monitoring with selfie vital signs and remote therapeutic monitoring allows expensive centralized clinical studies to be changed to decentralized outpatient studies
- Less expensive clinical studies with improved pharmacokinetic and pharmacodynamic data by avoiding patient logs

Status with iPill

- iPill will do a commercial pilot with a CRO in 2025
- iPill gets paid upfront by the CROs

iPill Team

CEO **John Hsu MD**
Co-Founder
Anesthesia & Pain

CFO **Rich Canote, MBA**
Engine Room LLC

CRO **Kendra Allen, BSN, MBA**
Insurance Fight Club inc
Billing/Collections

CSO **Vinay Gokhale, BS, MBA**
Thirdwayv inc

Board

John Hsu MD,
Co-Founder

Arthur Becker
Madison Advisors

Michele Leonhart BA
Former Director DEA
U.S. Justice Dept.
Drug Enforcement Agency

Tobias Fischli, MBA
Fischli Family Office

Robert Saunders, MBA
OCA Ventures

Special Advisors

Brian Harvey MD, PhD
Former FDA
Division Director

Kathleen O'Toole BA, JD, PhD
Former Police Commissioner
Boston and Seattle

Peter Weinstein PhD, JD
Former Baxter Counsel
Patent Prosecutor

Lynn Webster MD,
Senior Fellow, Center for U.S. Policy, Past President American Academy of Pain Medicine

iPill Future Vision and Management Strategy

Future

Growth

Other Drugs
Sedatives, Tranquilizers, Stimulants

TAM Sedatives, Tranquilizers, Stimulants
901.2 million prescriptions annually = $315billion $350/iPill
18.6 million used sedatives
39.3 million used tranquilizers
17.2 million used stimulants

Other countries
Canada, EU, UK

TAM CANADA 351,000 lives annually = $1.474 billion at $350/iPill
Canada 351,000 people with OUD per year

TAM UK 133,704 lives annually = $561.5 million at $350/iPill
UK 133,704 has people with OUD per year

A Win-Win-Win Value Proposition

Patients
Improved access to OUD care
Safe treatment of pain
Saved lives
Reduced medical expenses
Improved care

Physicians
Improved patient compliance
Reduced liability
Improved care
Monitor medication adherence

Investors
Social impact investment
$10 billion market

Pharmacies
Reduced liability
Reduced opioid litigation

Opioid suppliers
Reduced liability
Reduced opioid litigation

OUD clinics
Reduced relapse
Reduced transportation cost
Increase patient capacity
More value based contracts

Public Health
Reduced opioid diversion
Reduced $696 billion spend on OUD treatment, crime, homelessness, unemployment

Insurers
Reduced medical expenses
Improved margins

Politicians
Saved people from OUD
Solved the opioid crisis

PBMs
New product to sell
Safeguard opioids

DEA
Real-time tracking of opioids prescribed and used
Increased supply chain security to reduce loss
Real-time monitoring of chain of custody

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Case Study

Pilot Location Durham, North, Carolina
PI: John Hsu MD

Pilot

6 patients on Suboxone SL 16mg BID were instructed to download the app from the google play store and connect to the secure dispenser. Subsequent self-dispensing of their meds required a faceprint at which time vitals were collected from the selfie and a selfie videotape of pill ingestion

Results

All patients

- downloaded the app and operated the dispenser from their own Android phone
- self-dispense their Suboxone dose BID for a period of 30 days.
- No tampering was noted.
- 2 patients forgot to take their medications but were reminded to take their medications after calls from family. No other missed doses were noted.
- No patients were seen in the emergency room or required additional provider interactions for withdrawal symptoms.

Conclusion

A biometric secured dispenser operated by a smartphone app for precise home dispensing of medications can be used to ensure correct patient ingestion of medications as prescribed. This system could be used to prevent medication abuse, diversion, or accidental overdose.

Remote medication adherence monitoring changes inaccurate patient reports of medication adherence to objective medication adherence data so physicians can assess and create individualized care plans to improve the quality healthcare and save money and lives.

Workflow

Active controlled dispensing

Patient

Start

Every 30 days clinician prescribes medications

Patient receives prefilled dispenser from pharmacy

Download app

Biometric Authentication Faceprint/Gov't ID

Patient videotapes self dispense of prescribed medications

Finish

After 30 days dispenser is recycled/ disposed of by patient

Secure Storage

Unused pills destroyed at 30 days or immediately on tampering

Remote medication adherence monitoring

Provider

Usage reporting Detection/remediation of missed daily medication doses

Real-time care team notifications

Clinician cloud portal

Patient Interventions to reduce relapse

Data collection on efficacy of treatment

Accountability

Text, email, call patient, family, friends, caregiver to remediate missed daily medication dose